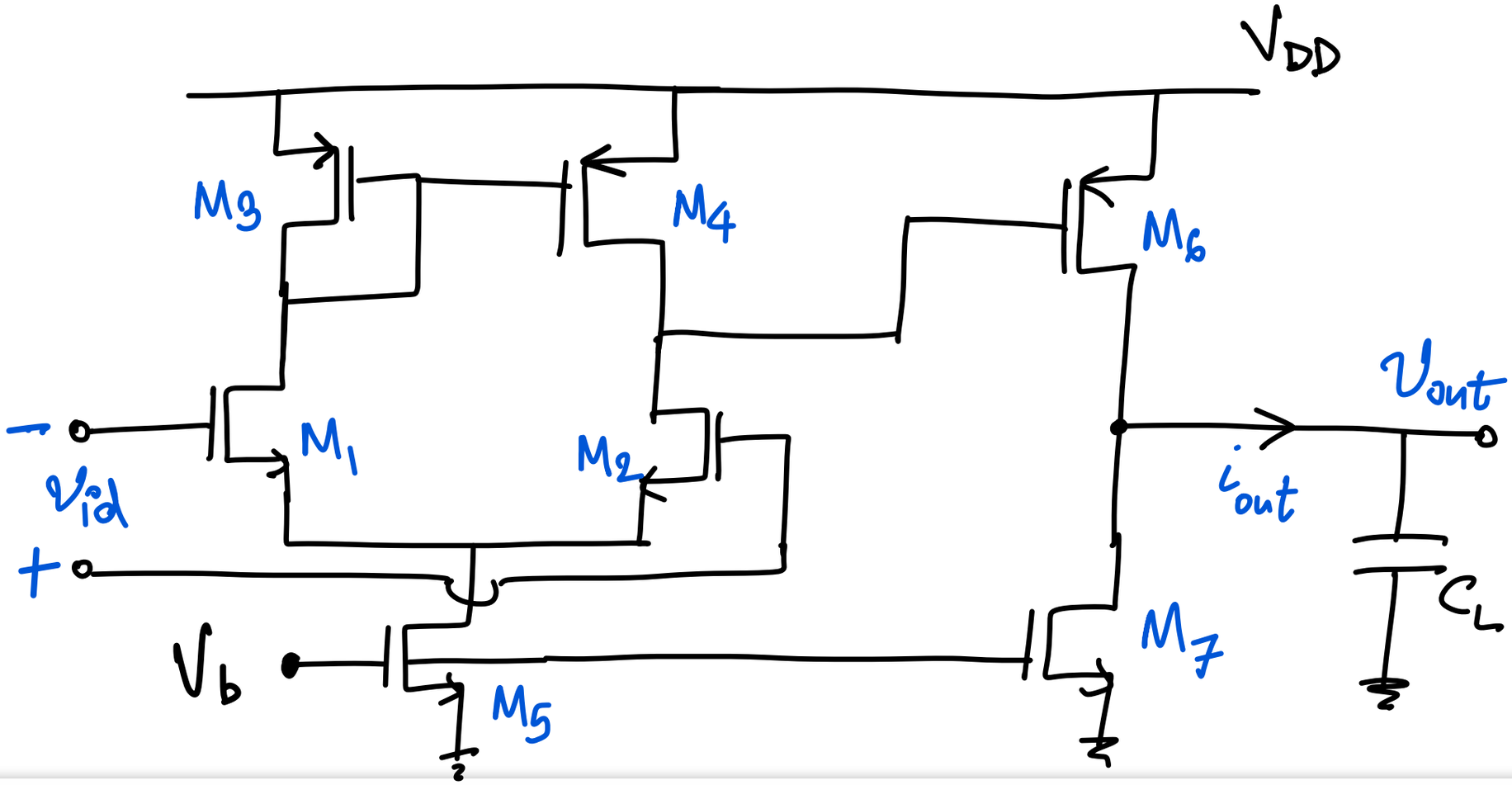


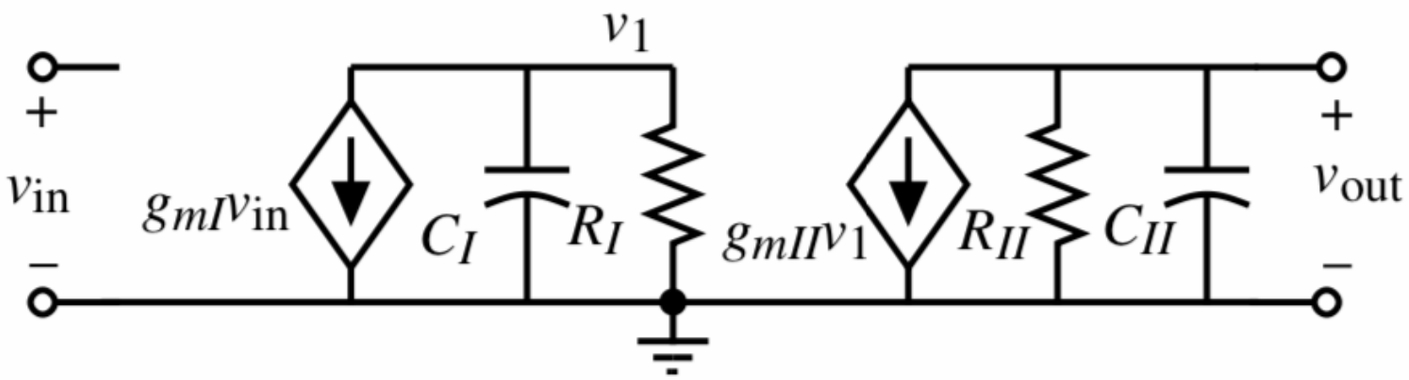
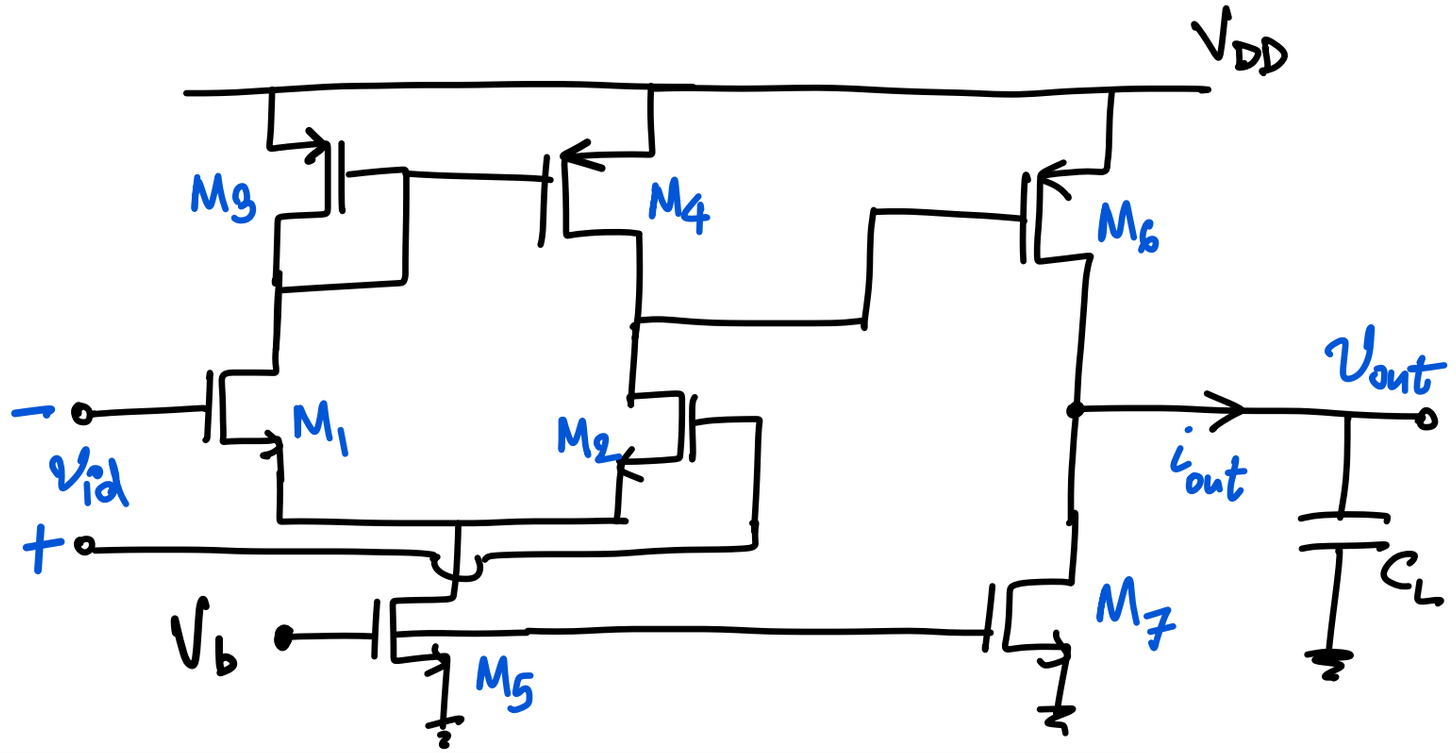


Figure 6.2-4 Second-order, small-signal equivalent circuit for a two-stage op amp.

$g_{m1} = g_{m1}$, $g_{mII} = g_{m6}$

$$A_{v0} = -g_{m1} \cdot g_{m6} \cdot R_I \cdot R_{II}, \quad p_I = \frac{-1}{R_I C_I}, \quad p_{II} = \frac{-1}{R_{II} C_{II}}$$

$$R_I = r_{o2} \| r_{o4}, \quad C_I = C_{DB4} + C_{DB2} + C_{SG6}, \quad R_{II} = r_{o6} \| r_{o7}, \quad C_{II} = C_L$$



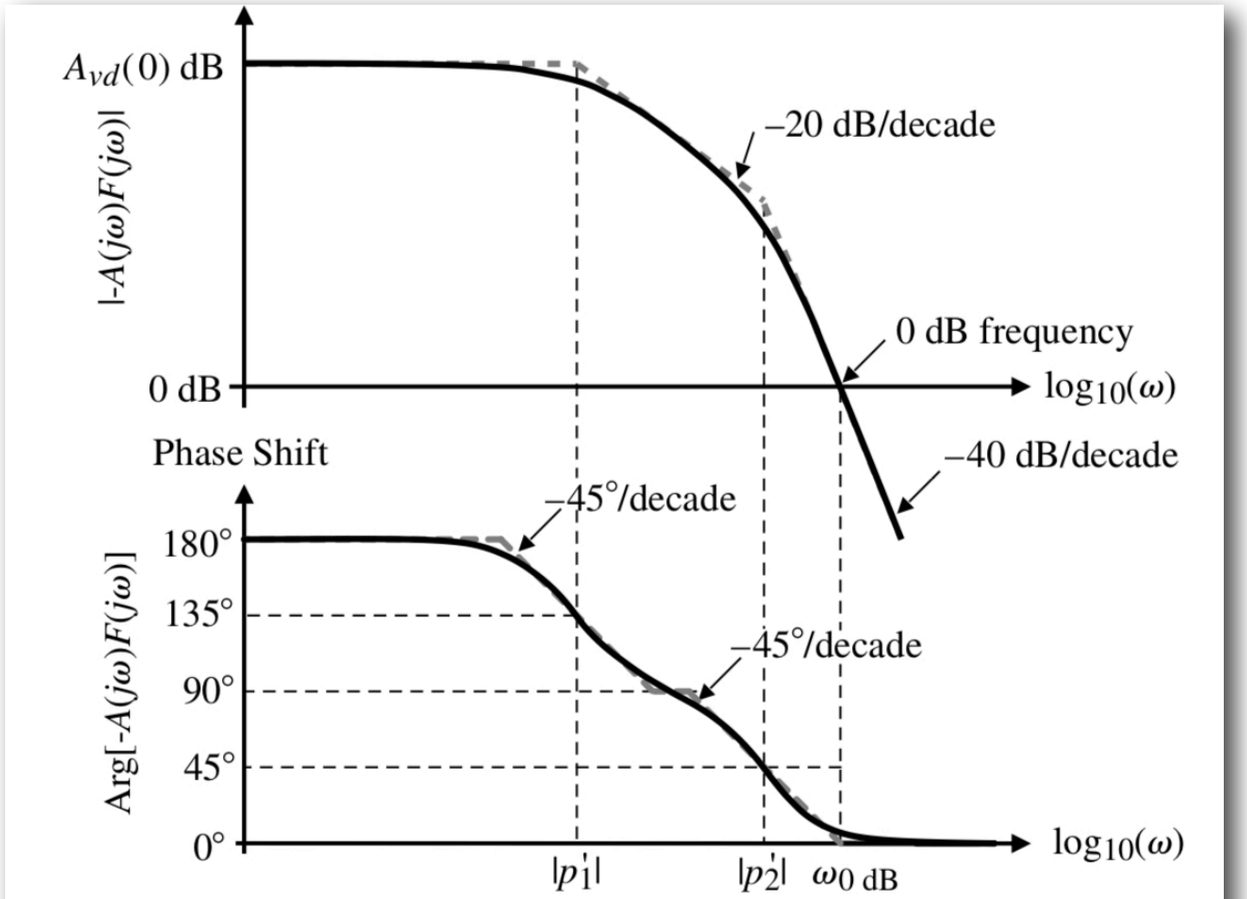
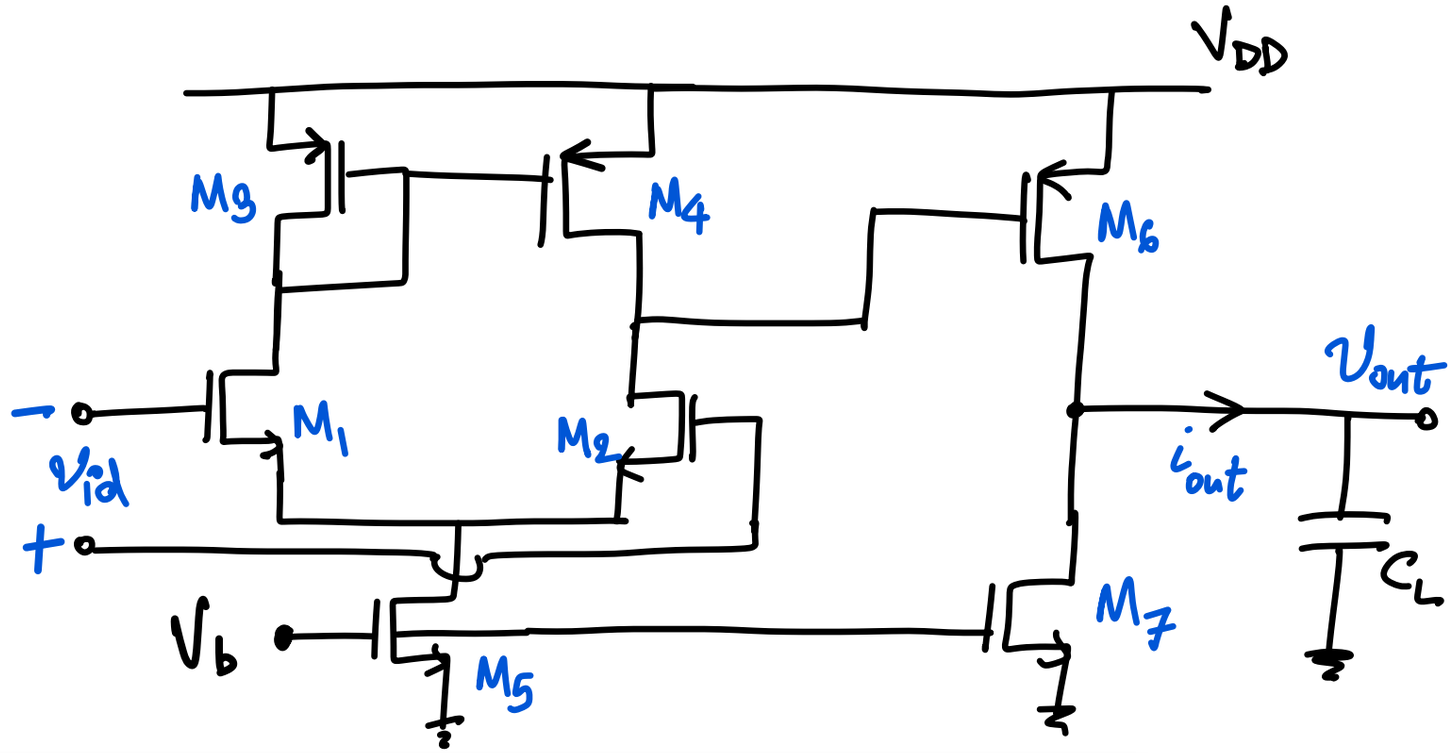


Figure 6.2-5 The open-loop frequency response of a negative-feedback loop using an uncompensated op amp and a feedback factor of $F(s) = 1$.

Uncompensated OTA

↓

Stability issue



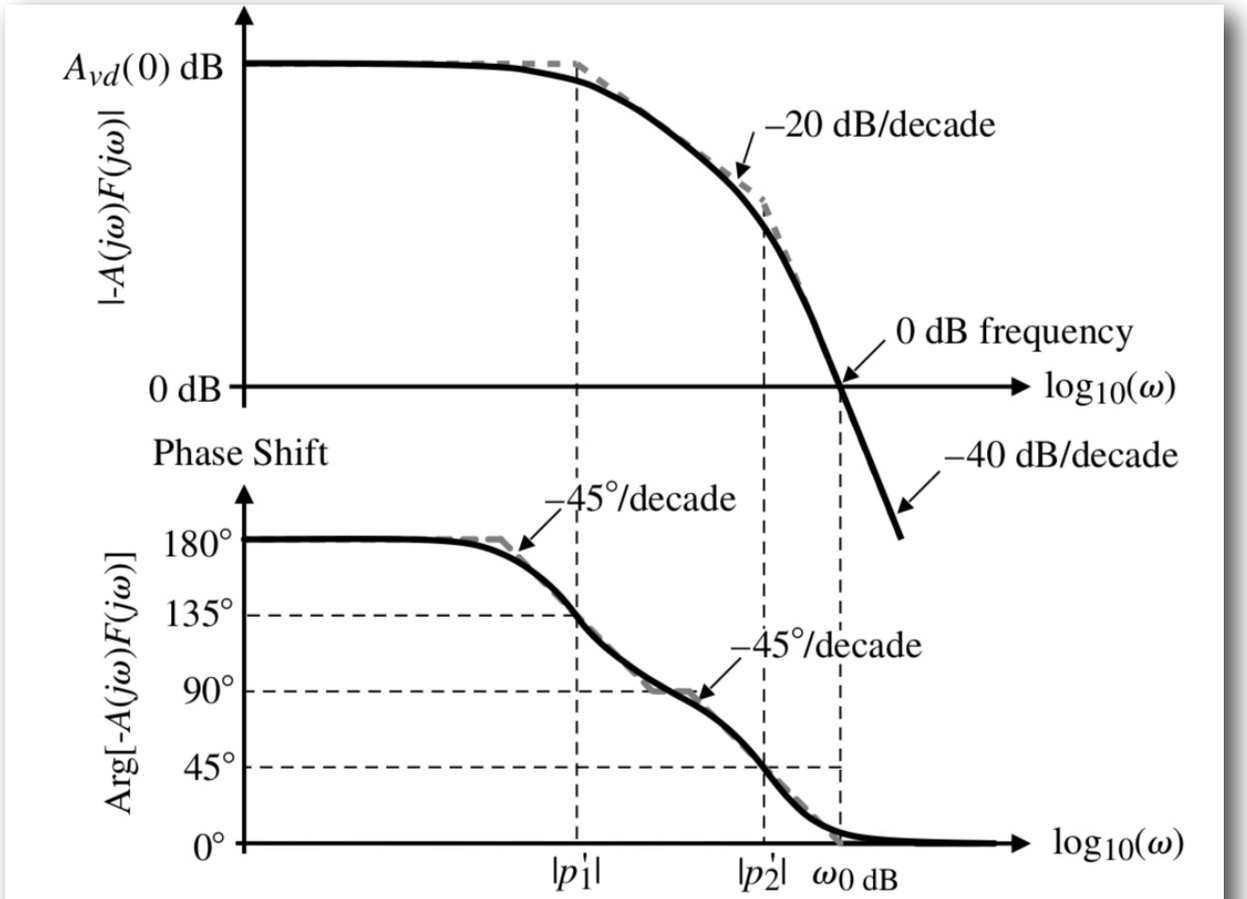


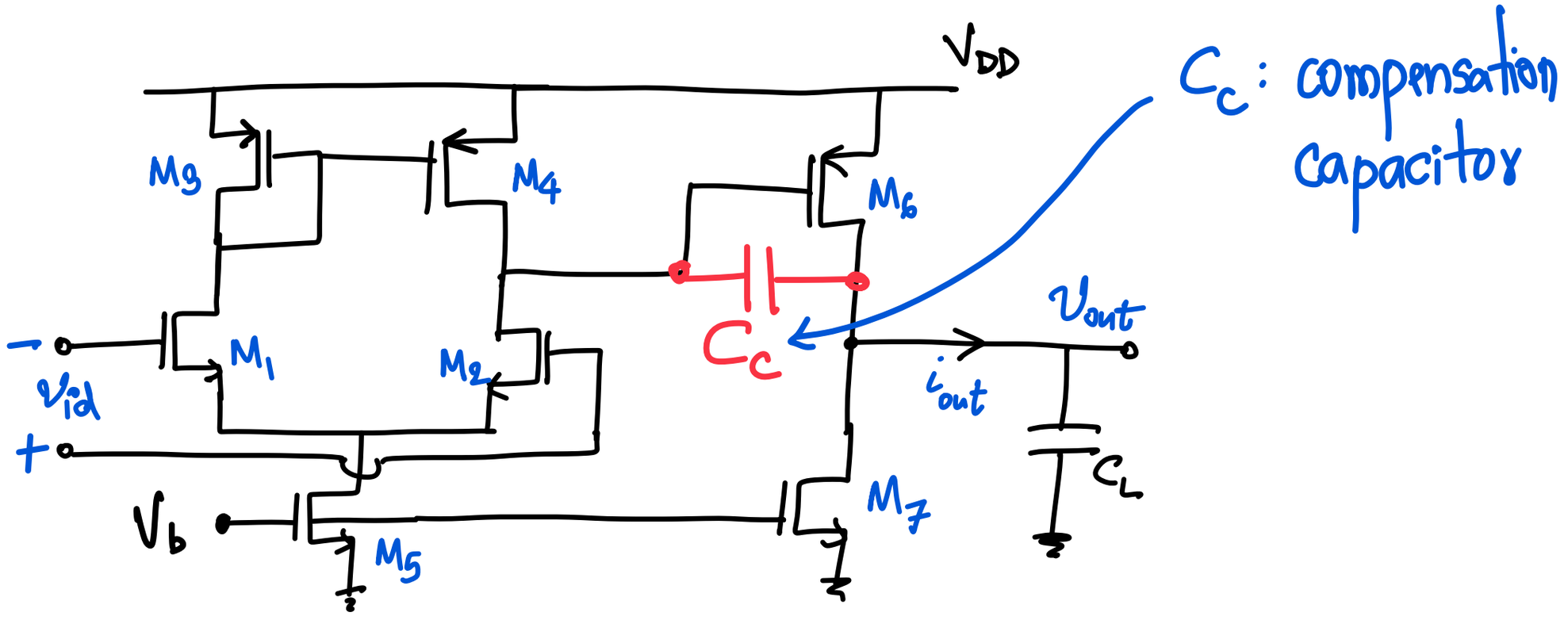
Figure 6.2-5 The open-loop frequency response of a negative-feedback loop using an uncompensated op amp and a feedback factor of $F(s) = 1$.

Uncompensated OTA

↓

Stability issue

Miller compensation of two-stage OTA



Miller compensation of two-stage OTA

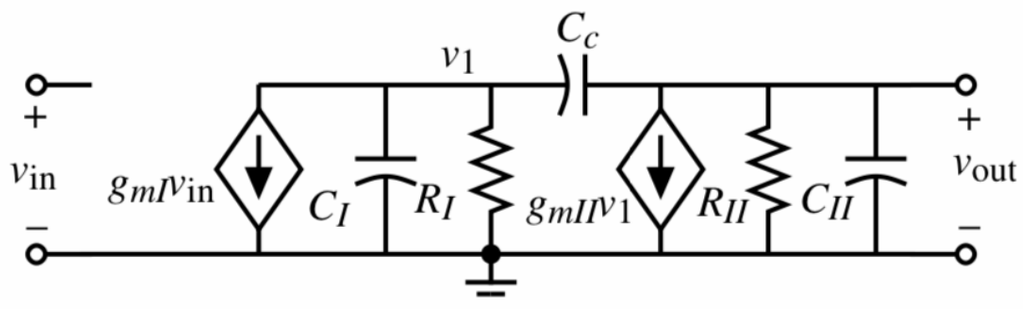


Figure 6.2-6 Miller capacitance applied to the two-stage op amp.

The overall transfer function that results from the addition of C_c is,

$$\frac{V_o(s)}{V_{in}(s)} = \frac{g_{m_I} \cdot g_{m_{II}} \cdot R_1 \cdot R_{II} \cdot (1 - s C_c/g_{m_{II}})}{1 + s[R_I(C_1 + C_c) + R_{II}(C_{II} + C_c) + g_{m_{II}} R_1 R_{II} C_c] + s^2 R_I R_{II}[C_1 C_{II} + C_c C_1 + C_c C_{II}]}$$

Ref: CMOS Analog Circuit Design by Allen & Holberg

Miller compensation of two-stage op-amp

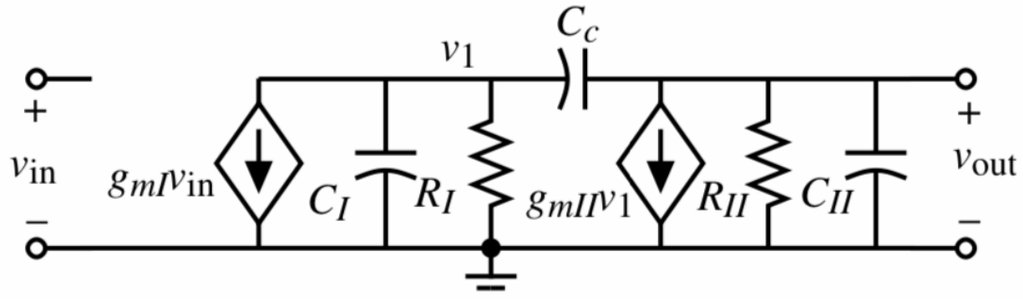


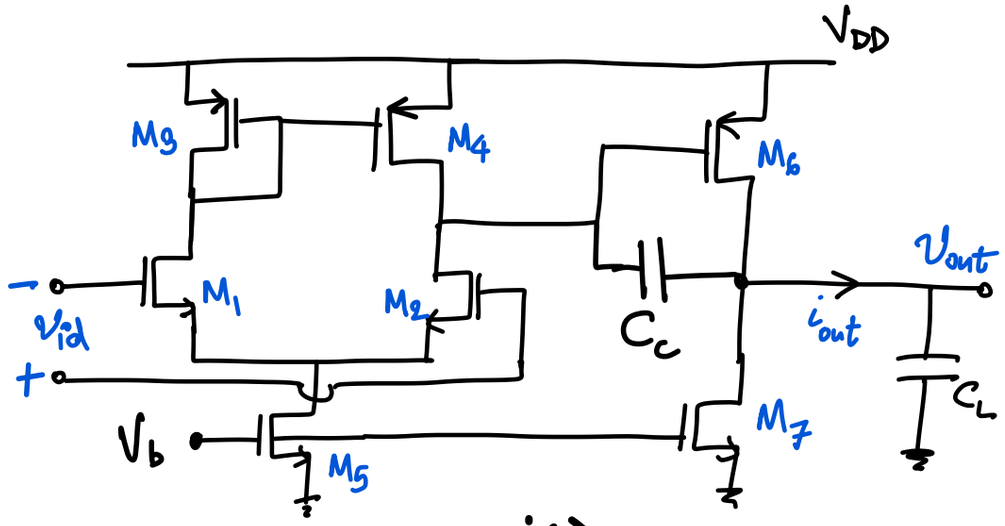
Figure 6.2-6 Miller capacitance applied to the two-stage op amp.

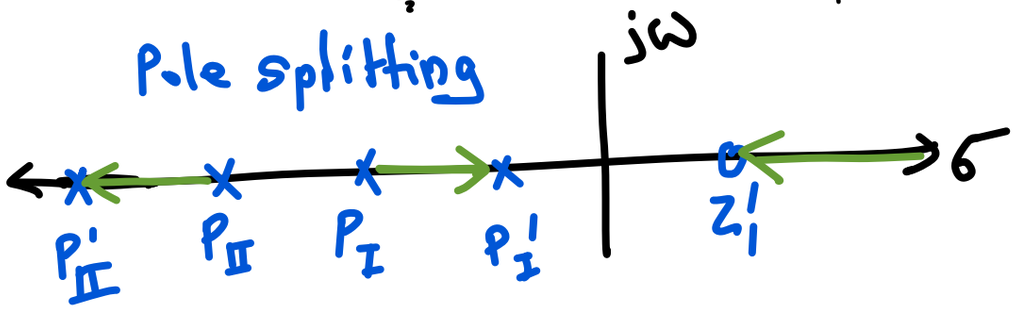
$$\frac{V_o(s)}{V_{in}(s)} = \frac{g_{m_I} \cdot g_{m_{II}} \cdot R_2 \cdot R_{II} \cdot (1 - s C_c/g_{m_{II}})}{1 + s[R_I(C_I + C_c) + R_{II}(C_{II} + C_c) + g_{m_{II}} R_2 R_{II} C_c] + s^2 R_I R_{II}[C_I C_{II} + C_c C_I + C_c C_{II}]}$$

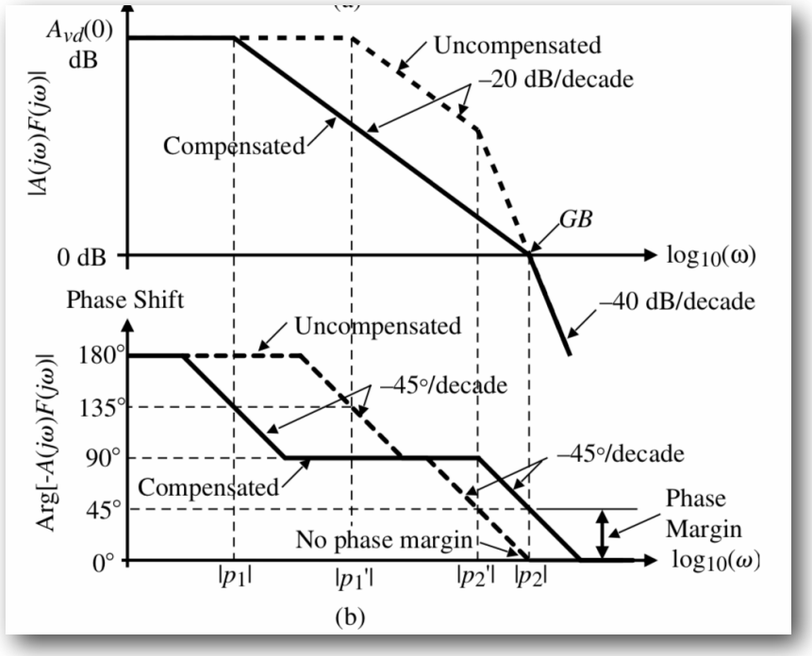
Assuming two widely spaced poles,

$$p_I' \approx \frac{-1}{g_{m_{II}} \cdot R_I \cdot R_{II} \cdot C_c} \; ; \; p_{II}' \approx \frac{-g_{m_{II}}}{C_{II}} \; ; \; z_1 = \frac{g_{m_{II}}}{C_c}$$

Pole splitting (an important concept)







$P_I = \frac{-1}{R_I C_I}$

$$P_I' = \frac{-1}{R_I \cdot g_{m_{II}} R_{II} \cdot C_c}$$

(Pole moves towards origin)

$P_{II} = \frac{-1}{R_{II} \cdot C_{II}}$

$$P_{II}' = -\frac{g_{m_{II}}}{C_{II}} \approx \frac{g_{m6}}{C_L}$$

(Pole moves away from origin)

$$\left[g_{m_{II}} R_{II} \gg 1 \Rightarrow g_{m_{II}} \gg \frac{1}{R_{II}} \right]$$

$Z_1' = \frac{g_{m_{II}}}{C_c}$

[No zero without C_c. $\Rightarrow Z_1 \to \infty$]